

A Distributed Economic Dispatch Strategy for Power–Water Networks

Peng Li, Jiangping Hu , *Senior Member, IEEE*, Li Qiu , *Fellow, IEEE*, Yiyi Zhao, and Bijoy Kumar Ghosh , *Fellow, IEEE*

Abstract—A power–water network consists of power plants, water plants, and coproduction plants, which supply power and water collectively. Heterogeneous plants generally have different costs when they produce water and power; thus, an optimization method should be employed to allocate the production tasks of the plants to reduce the total cost of the power–water network. This study investigates a continuous-time distributed algorithm for the economic dispatch problem of the power–water network. A conceptual model is first presented to illustrate the logical relationships within the power–water network, and then, a constrained nonlinear programming model is developed to describe the economic dispatch problem of the power–water network. The model not only includes the generation capacity constraints and supply–demand constraints but also considers the constraints on the power–water production ratios of the coproduction plants. A novel distributed algorithm is developed for the economic dispatch problem, which can guarantee a plug-and-play operation and preserve the privacy information of the plants in the power–water network. Some numerical results are given to demonstrate the performance and effectiveness of the proposed algorithm.

Index Terms—Distributed optimization, economic dispatch problem, nonsmooth cost function, power–water network.

I. INTRODUCTION

AS WE know, power and water are essential resources for the development of a society. Furthermore, these two resources are closely related in production, distribution, consumption, etc. For example, a power system requires a large amount of water for power supply (e.g., coal exploitation and steam–power plant), and a large amount of power is required to operate water facilities (multistage flash distillation, pumping, waste water treatment, etc.). This interlinked power–water system is also called the energy–water nexus [1], which can be illustrated in Fig. 1.

In this study, we assume that water and coproduction plants have their own water sources. Moreover, we do not consider the power and water storage facilities in the problem model, because the power storage facilities have such small storage capacity that their influence can be neglected. On the contrary, the volume of water storage facilities is often high; water can be released frequently to satisfy the demand both in production and daily life. It can be concluded that we do not need to consider the storage volume of water storage plants during a small period. Instead, the water storage plants can belong to the set of water production plants. Under these assumptions, we give a mathematical model of the economic dispatch problem for the power–water network. The objective of this model is to find an optimal solution, which makes the whole production cost reach minimum for the network. Then, the economic dispatch problem is formulated as follows:

$$\begin{aligned}
 \min \quad & \sum_{i=1}^{n_p} C_{pi}(x_{pi}) + \sum_{j=1}^{n_w} C_{wj}(x_{wj}) \\
 & + \sum_{k=1}^{n_c} C_{ck}(x_{cpk}, x_{cwk}) \\
 \text{s.t.} \quad & \underline{x}_{pi} \leq x_{pi} \leq \bar{x}_{pi} \text{ for } i = 1, \dots, n_p \\
 & \underline{x}_{wj} \leq x_{wj} \leq \bar{x}_{wj} \text{ for } j = 1, \dots, n_w \\
 & \underline{x}_{cpk} \leq x_{cpk} \leq \bar{x}_{cpk} \\
 & \underline{x}_{cwk} \leq x_{cwk} \leq \bar{x}_{cwk} \text{ for } k = 1, \dots, n_c
 \end{aligned} \tag{1}$$

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Peng Li is with the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: pengli_530@163.com).

Jiangping Hu is with the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: hujp@uestc.edu.cn).

Li Qiu is with the Department of Electronic and Computer Engineering, The Hong Kong University of Science and Technology, Kowloon, Hong Kong (e-mail: eeqiu@ust.hk).

Yiyi Zhao is with the School of Business Administration, Faculty of Business Administration, Southwestern University of Finance and Economics, Chengdu 611130, China (e-mail: zhaoyi@swufe.edu.cn).

Bijoy Kumar Ghosh is with the Department of Mathematics and Statistics, Texas Tech University, Lubbock, TX 79409 USA, and also with the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: bijoy.ghosh@ttu.edu).

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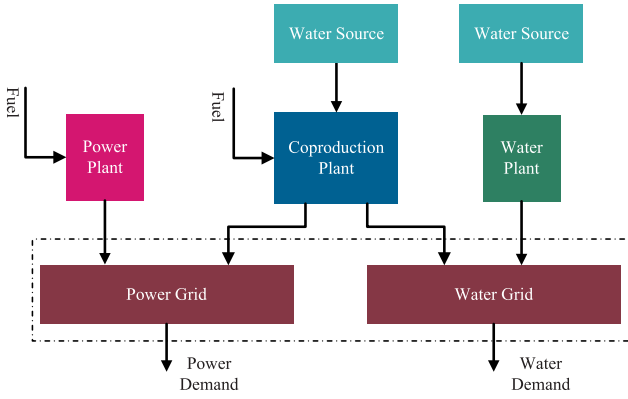


Fig. 1. Power–water system.

$$\sum_{i=1}^{n_p} x_{pi} + \sum_{k=1}^{n_c} x_{cpk} = \sum_{i=1}^{n_p} d_{pi} + \sum_{k=1}^{n_c} d_{cpk} \quad (2)$$

$$\sum_{j=1}^{n_w} x_{wj} + \sum_{k=1}^{n_c} x_{cwk} = \sum_{j=1}^{n_w} d_{wj} + \sum_{k=1}^{n_c} d_{cwk} \quad (3)$$

$$\underline{r}_k \leq \frac{x_{cpk}}{x_{cwk}} \leq \bar{r}_k \text{ for } k = 1, \dots, n_c \quad (4)$$

where $x_{pi} \in \mathbb{R}$ and $x_{wj} \in \mathbb{R}$ denote the power generated by the power plant at node i and the water produced by the water plant at node j , respectively; $x_{cpk} \in \mathbb{R}$ and $x_{cwk} \in \mathbb{R}$ denote the power and water produced by the coproduction plant at node k , respectively; d_{pi} denotes the local power demand of node i , d_{wj} denotes the local water demand of node j , and d_{cpk} and d_{cwk} denote the local power demand and the local water demand of node k , respectively; $C_{pi}(x_{pi})$ is the cost function of the i th power plant; $C_{wj}(x_{wj})$ is the cost function of the j th water plant; and $C_{ck}(x_{cpk}, x_{cwk})$ is the cost function of the k th coproduction plant, respectively. In addition, n_p , n_w , and n_c denote the number of power plants, water plants, and coproduction plants, respectively. The inequality constraints (1) represent the production capacity of plants, where $\underline{x}_{pi}$, $\bar{x}_{pi}$, $\underline{x}_{wj}$, $\bar{x}_{wj}$, $\underline{x}_{cpk}$, $\bar{x}_{cpk}$, $\underline{x}_{cwk}$, and $\bar{x}_{cwk}$ are nonnegative real numbers. Equalities (2) and (3) denote the supply–demand balance of power and water, respectively. Additionally, the constants $\underline{r}_k$ and $\bar{r}_k$ in the inequality constraints (4) are the upper and lower bounds on the power–water production ratio of the coproduction plant at node k .

The economic dispatch problem of the power–water system aims at minimizing the production cost of power and water while satisfying the constraint of production capacity and the supply–demand balance condition. Many algorithms for the economic dispatch problem, such as genetic algorithm [2] and particle swarm method [3], have been developed. In most of the aforementioned references, these approaches generally need a centralized unit to collect the data of the entire network, compute the optimal solution, and send it to all the plants. However, these approaches may not be suitable for the economic dispatch problem in the large-scale network due to the external disturbances, unreliability of information channel, privacy concerns, and the

failure of devices [4], [5]. Motivated by the research of multi-agent system [6]–[11], the distributed optimization algorithm may be a good alternative for this class of problems.

For the distributed optimization algorithms, all the agents in the system cooperatively seek the optimal decisions by locally processing the private data and the information from their neighbors. To eliminate the requirement of the central unit, various distributed optimization algorithms [12]–[24] have been developed for economic dispatch problems of power networks. The seminal studies [12]–[14] mainly considered the relaxation of economic dispatch problem, that is, the agent’s local constraints are not considered. However, the relaxed problem model is not practical in most applications. For example, in the economic dispatch problem of the power system, the local box constraint, which represents the capacity of generators, must be considered for the system normal operation. To deal with the local box constraint, many distributed economic dispatch algorithms [15]–[19] have been designed based on the optimization conditions of the relaxed problems in [13]. In particular, the authors of [15] and [16] employed these conditions to study the economic dispatch problem, which assumes that the local cost functions are quadratic. Yang *et al.* [17] adopted the gradient push-sum method to design the distributed economic dispatch algorithm, where the local cost functions are assumed to be strictly convex. The authors of [18] and [19] relaxed the assumption from strictly cost functions to convex cost functions and adopted the exact penalty function method to deal with the local box constraint.

Furthermore, the authors of [20]–[24] adopted the (differentiated) projection operation to design some distributed algorithms for the economic dispatch problem involving the local set constraint. As described above, the distributed algorithms for the economic dispatch problem have been well studied, but the requirements from industrial production, such as the secure operation of the coproduction plants in power–water system, promote the research on the distributed economic dispatch problem with the local inequality constraints. Theoretically, the local inequality constraints can be converted into a set constraint, and the corresponding optimization problem should be solved by the algorithms [20]–[24]. However, when the projection operator is used to deal with the general inequality constraint, it is hard to obtain the analytical expression of the projected value. For this case, an optimization problem should be solved to obtain the projected value, which makes the projection operator difficult to deal with. Recently, a few papers on distributed optimization [25]–[27] have considered the local inequality constraints; however, they are not suitable for the optimization problem with a coupled equality constraint. Therefore, it is necessary to develop novel distributed algorithms for the optimization problem with the coupled equality constraint and the local nonlinear inequality constraint.

In this article, a continuous-time distributed algorithm is developed for an economic dispatch problem of a power–water network. In the network, each plant or agent optimizes its own cost function and exchanges the information with its neighbors over a communication network. The main contributions of this study are provided as follows: First, for the power–water network, we formulate a new economic dispatch problem, which

includes not only supply–demand constraints and individual generator limits but also power–water production ratio constraints. Second, the cost functions are assumed to be convex and can even be nonsmooth. Moreover, the proposed dispatch algorithm can globally converge to the optimal solution from any initial state. Finally, the proposed algorithm is fully distributed, which implies that each agent in the multiagent system only exchanges the information with its neighbors, and thus, the privacy of agents can be favorably protected.

The structure of this note is given as follows. First, some results on algebraic graph theory, nonsmooth analysis, set-valued dynamical systems, and the hybrid economic dispatch problem are provided in Section II. Subsequently, a continuous-time distributed algorithm is proposed and its convergence analysis is completely given in Section III. In Section IV, a specific case is given to validate the proposed algorithm. Finally, Section V concludes this article.

Notations: $\mathbb{R}$ represents the set of real numbers, $\mathbb{R}_+$ denotes the set of nonnegative real numbers, and $\mathbb{R}^n$ represents the set of n -dimensional real vectors. $\mathbf{x}^T$ represents the transpose of the column vector $\mathbf{x}$, x_i is the i th element of vector $\mathbf{x}$, and $\text{col}(x_1, \dots, x_n) = (x_1^T, \dots, x_n^T)^T$. Denote $\mathbf{1} = (1, \dots, 1)^T$ and $\mathbf{0} = (0, \dots, 0)^T$ with appropriate dimensions based on the context. I_n is the $n \times n$ identity matrix. The inequality symbol $\leq$ denotes inequality between scalars or componentwise inequality between vectors. The Euclidean norm of a vector is denoted by $\|\cdot\|$. The Kronecker product of matrices A and B is denoted by $A \otimes B$. Define $B(x, \delta) = \{y \in \mathbb{R}^n : \|y - x\| < \delta\}$. Moreover, define a particular operator $\mu^+ = \text{col}(\mu_1^+, \mu_2^+, \dots, \mu_n^+)$, where $\mu_i^+ = \max\{\mu_i, 0\}$, for $i = 1, \dots, n$.

II. PRELIMINARIES AND PROBLEM FORMULATION

In this section, we introduce some basic concepts and notations on algebraic graph theory, nonsmooth analysis and differential inclusion, and the hybrid economic dispatch problem.

A. Algebraic Graph Theory

An undirected graph G consists of the node set $V = \{1, \dots, n\}$, the edge set $E \subseteq V \times V$, and the adjacency matrix $A \in \mathbb{R}^{n \times n}$ [28]. If the node i can receive information from node j , then there exists an edge from node i to node j , denoted by e_{ij} . The element a_{ij} of the adjacency matrix A is positive if the edge e_{ij} exists, and $a_{ij} = 0$, otherwise. A graph is undirected if $a_{ij} = a_{ji}$, for all $i, j \in \{1, \dots, n\}$. The neighbor set of node i is defined as $N_i = \{j \in V : a_{ij} \neq 0\}$. If there exists a path between any pair of nodes, then the undirected graph is connected. The degree of the node i is given by $d_i = \sum_{j=1}^n a_{ij}$, and the Laplacian matrix associated with the graph G is defined by $L = D - A$, where $D = \text{diag}\{d_1, \dots, d_n\}$ is the degree matrix. If the graph G is an undirected and connected graph, then 0 is a simple eigenvalue of L and the other eigenvalues are positive. Moreover, $L\mathbf{x} = \mathbf{0}$ if $x_1 = \dots = x_n$. Define an

orthogonal matrix $[r \ R] \in \mathbb{R}^{n \times n}$, where

$$\begin{aligned} r &= \frac{1}{\sqrt{n}}\mathbf{1}, \quad r^T R = \mathbf{0}^T \\ R^T R &= I_{n-1}, \quad R R^T = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T \\ \begin{bmatrix} r^T \\ R^T \end{bmatrix} L \begin{bmatrix} r & R \end{bmatrix} &= \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & R^T L R \end{bmatrix}. \end{aligned} \quad (5)$$

B. Nonsmooth Analysis and Differential Inclusion

We introduce some results from nonsmooth analysis [29]. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is locally Lipschitz at $x \in \mathbb{R}^n$ if there exist $\epsilon, \delta \in (0, +\infty)$ such that $\|f(y) - f(z)\| \leq \epsilon\|y - z\|$ for any $y, z \in B(x, \delta)$. A function f is said to be locally Lipschitz in $\mathbb{R}^n$ if it is locally Lipschitz at any $x \in \mathbb{R}^n$. Moreover, f is globally Lipschitz on $\mathbb{R}^n$ if for all $y, z \in \mathbb{R}^n$, there exists $C \in \mathbb{R}_+$ such that $\|f(y) - f(z)\| \leq C\|y - z\|$. Assume that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is locally Lipschitz in $\mathbb{R}^n$; let $\Omega_f \subset \mathbb{R}^n$ denote the set of points, where f is not differentiable, and then, the generalized gradient ∂f is defined as

$$\partial f(x) = \text{co} \left\{ \lim_{k \rightarrow +\infty} \nabla f(x_k) : x_k \rightarrow x, x_k \notin \Omega_f \cup S \right\}$$

where $S \subset \mathbb{R}^n$ is a set of measure zero and co denotes convex hull. The generalized gradient ∂f of locally Lipschitz function f is nonempty, compact, convex, upper semicontinuous, and locally bounded at x .

A differential inclusion is defined as

$$\dot{x}(t) \in F(x(t)) \quad (6)$$

where $t \in \mathbb{R}_+$ and $F(x(t)) : \mathbb{R}^n \rightarrow \mathfrak{B}(\mathbb{R}^n)$ is a set-valued map, where $\mathfrak{B}(\mathbb{R}^n)$ denotes the collection of all subsets of $\mathbb{R}^n$. The set of equilibrium points for differential inclusion (6) is defined as $S_e = \{x : 0 \in F(x)\}$. A Caratheodory solution of (6) on $[t_0, t_1] \in [0, +\infty)$ is an absolutely continuous map $x : [t_0, t_1] \rightarrow \mathbb{R}^n$ such that $\dot{x}(t) \in F(x(t))$ for almost every $t \in [t_0, t_1]$. A set M is said to be weakly invariant (respectively, strongly invariant) with respect to (6), if for every $x \in M$, at least a solution (respectively, all the solutions) of (6) starting from x is entirely contained in M . If the function $V(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable, the set-valued Lie derivative of V with respect to differential inclusion (6) at x is defined as $\mathcal{L}_F V(x) = \{(\nabla V(x))^T v : v \in F(x)\}$. Next, the LaSalle invariance principle for the differential inclusion (6) is given as follows [29].

Lemma 2.1: For the differential inclusion (6), we assume that F is upper semicontinuous and locally bounded, and $F(x)$ takes nonempty, compact, and convex values. Let $V : \mathbb{R}^q \rightarrow \mathbb{R}$ be a locally Lipschitz and regular function, $S \subset \mathbb{R}^q$ be compact and strongly invariant for (6), $\phi(\cdot)$ be a solution of (6),

$$\mathcal{R} = \{x \in \mathbb{R}^q : 0 \in \mathcal{L}_F V(x)\} \quad (7)$$

and M be the largest weakly invariant subset of $\overline{\mathcal{R}} \cap S$, where $\overline{\mathcal{R}}$ is the closure of $\mathcal{R}$. If $\max \mathcal{L}_F V(x) \leq 0$ for all $x \in S$, then $d(\phi(t), M) \rightarrow 0$ as $t \rightarrow +\infty$, where $d(\phi(t), M) = \inf\{\|\phi(t) - y\|, y \in M\}$.

C. Hybrid Economic Dispatch

Although the nonlinear programming model (1)–(4) clearly describes the economic dispatch problem for the supply side of the power–water network, for the convenience of theoretical analysis, we further rewrite the model as a general nonlinear programming model as follows:

$$\begin{aligned} \min f(x) &= \sum_{i=1}^n f_i(x_i) \\ \text{st. } \sum_{i=1}^n x_i &= \sum_{i=1}^n d_i \\ g_i^j(x_i) &\leq 0, \text{ for } j = 1, \dots, J^i, i = 1, \dots, n \end{aligned} \quad (8)$$

where $x_i \in \mathbb{R}^m$ denotes the allocation variable of plant i , $x = \text{col}(x_1, \dots, x_n)$ is the allocation vector of the whole system, $f_i(x_i) : \mathbb{R}^m \rightarrow \mathbb{R}$ denotes the cost function of plant i , $d_i \in \mathbb{R}^m$ is the local demand, and $g_i^j(x_i) : \mathbb{R}^m \rightarrow \mathbb{R}$ ($j = 1, \dots, J^i$) denote the inequalities described by the local constraint (1) or (4). For the nonlinear programming problem (8), we give the following assumptions.

Assumption 2.2: There exists a solution x_i such that $\sum_{i=1}^n x_i = \sum_{i=1}^n d_i$ and $g_i^j(x_i) < 0$.

Assumption 2.3: The cost functions $f_i(x_i)$ and the constraint functions $g_i^j(x_i)$ are Lipschitz and convex on $\mathbb{R}^m$.

Under Assumption 2.2, the Slater's constraint condition is satisfied [30]. Assumption 2.3 implies that f is convex on the set $X = \{x : \sum_{i=1}^n x_i = \sum_{i=1}^n d_i, g_i^j(x_i) \leq 0 \forall j, i\}$, and thus, there exists a solution to problem (8). The following assumption on graph G is also standard in distributed optimization [20], [21], [31], and it is mainly used to guarantee that each agent's information can be received by the other agents in the system.

Assumption 2.4: The graph $G = (V, E, A)$ is not only undirected but also connected.

From [30, Th. 3.34], it is straightforward to obtain the following optimization conditions for the problem (8).

Lemma 2.5: Under Assumptions 2.2 and 2.3, x^* is a solution to the optimization problem (8) if and only if there exist v^* and $\mu_{ij}^* \geq 0$, for $j = 1, \dots, J^i$, $i = 1, \dots, n$ satisfying the following conditions:

$$0 \in \partial f_i(x_i^*) + \sum_{j=1}^{J^i} \mu_{ij}^* \partial g_i^j(x_i^*) + v^* \quad \forall i \quad (9a)$$

$$g_i^j(x_i^*) \leq 0, \mu_{ij}^* \geq 0, \mu_{ij}^* g_i^j(x_i^*) = 0 \quad \forall j, i \quad (9b)$$

$$\sum_{i=1}^n x_i^* = \sum_{i=1}^n d_i. \quad (9c)$$

Based on Lemma 2.5, the optimization objective turns to find the point x^* with Lagrangian multipliers v^* and μ_{ij}^* in a distributed manner. Next, an equivalent condition is given for the condition (9b), which is essential in the design of a distributed algorithm.

Lemma 2.6: $g_i^j(x_i) : \mathbb{R}^m \rightarrow \mathbb{R}$ and $\mu_{ij} \in \mathbb{R}$ satisfy the complementary slackness condition as follows:

$$g_i^j(x_i) \leq 0, \mu_{ij} \geq 0, \mu_{ij} g_i^j(x_i) = 0 \quad (10)$$

if and only if

$$\mu_{ij} = (\mu_{ij} + g_i^j(x_i))^+. \quad (11)$$

Proof: If μ_{ij} and $g_i^j(x_i)$ satisfy (11), then one has $\mu_{ij} = 0$ or $\mu_{ij} = \mu_{ij} + g_i^j(x_i) \geq 0$. For $\mu_{ij} = 0$, we have $g_i^j(x_i) \leq 0$ and $\mu_{ij} g_i^j(x_i) = 0$. When $\mu_{ij} = \mu_{ij} + g_i^j(x_i) \geq 0$, we have $g_i^j(x_i) = 0$, $\mu_{ij} \geq 0$ and $\mu_{ij} g_i^j(x_i) = 0$. Hence, we can conclude that $g_i^j(x_i) \leq 0$, $\mu_{ij} \geq 0$, $\mu_{ij} g_i^j(x_i) = 0$, when $\mu_{ij} = (\mu_{ij} + g_i^j(x_i))^+$.

Conversely, we assume that the complementary slackness condition (10) holds. It is easy to obtain that the complementary slackness condition (10) consists of two cases. One case is $\mu_{ij} = 0$, $g_i^j(x_i) \leq 0$, and the other case is $\mu_{ij} > 0$, $g_i^j(x_i) = 0$. When $\mu_{ij} = 0$, $g_i^j(x_i) \leq 0$, we have $\mu_{ij} = 0$, $\mu_{ij} + g_i^j(x_i) \leq 0$, and thus, $\mu_{ij} = (\mu_{ij} + g_i^j(x_i))^+$. When $\mu_{ij} > 0$ and $g_i^j(x_i) = 0$, we have $\mu_{ij} = \mu_{ij} + g_i^j(x_i) > 0$, and therefore, $\mu_{ij} = (\mu_{ij} + g_i^j(x_i))^+$. This completes the proof. $\square$

III. DISTRIBUTED ALGORITHM FOR THE ECONOMIC DISPATCH PROBLEM

We now provide a distributed algorithm for the nonlinear programming problem (8) as follows:

$$\begin{cases} \dot{x}_i \in -\partial f_i(x_i) - \lambda_i - \rho \left(x_i - d_i - \sum_{j \in N_i} (z_i - z_j) \right) \\ \quad - \sum_{j=1}^{J^i} \hat{\mu}_{ij} \partial g_i^j(x_i) \\ \dot{\mu}_{ij} = -\mu_{ij} + (\mu_{ij} + g_i^j(x_i))^+, j = 1, \dots, J^i \\ \dot{\lambda}_i = -\sum_{j \in N_i} (\lambda_i - \lambda_j) + \left(x_i - d_i - \sum_{j \in N_i} (z_i - z_j) \right) \\ \dot{z}_i = x_i - d_i - \sum_{j \in N_i} (z_i - z_j) \end{cases} \quad (12)$$

where $\mu_{ij} \in \mathbb{R}$, $\lambda_i \in \mathbb{R}^m$, and $z_i \in \mathbb{R}^m$. Moreover, $\hat{\mu}_{ij} = (\mu_{ij} + g_i^j(x_i))^+$, ρ is a positive constant. Let $x = \text{col}(x_1, \dots, x_n) \in \mathbb{R}^{nm}$, $\lambda = \text{col}(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^{nm}$, $z = \text{col}(z_1, \dots, z_n) \in \mathbb{R}^{nm}$, $d = \text{col}(d_1, \dots, d_n) \in \mathbb{R}^{nm}$ and $\mathfrak{S}_\mu = \text{col}(\sum_{j=1}^{J^1} \hat{\mu}_{1j} \partial g_1^j(x_1), \dots, \sum_{j=1}^{J^n} \hat{\mu}_{nj} \partial g_n^j(x_n)) \in \mathbb{R}^{nm}$. Moreover, define $\mu = \text{col}(\mu_1, \dots, \mu_n) \in \mathbb{R}^{\sum_{i=1}^n J^i}$ and $g(x) = \text{col}(g_1(x_1), \dots, g_n(x_n)) \in \mathbb{R}^{\sum_{i=1}^n J^i}$, where $\mu_i = \text{col}(\mu_{i1}, \dots, \mu_{iJ^i}) \in \mathbb{R}^{J^i}$ and $g_i = \text{col}(g_i^1, \dots, g_i^{J^i}) \in \mathbb{R}^{J^i}$. Then, the distributed algorithm (12) can be rewritten in a compact form as follows:

$$\begin{cases} \dot{x} \in -\partial f(x) - \lambda - \rho(x - d - (L \otimes I_m)z) - \mathfrak{S}_\mu \\ \dot{\mu} = -\mu + (\mu + g(x))^+ \\ \dot{\lambda} = -(L \otimes I_m)\lambda + (x - d - (L \otimes I_m)z) \\ \dot{z} = x - d - (L \otimes I_m)z. \end{cases} \quad (13)$$

Remark 3.1: Even though the proposed algorithm (12) is mainly employed to solve the economic dispatch problem (8) of the power–water network with a 2-D decision state, in fact,

it can be used to solve the optimization problem (8) in an m -dimensional Euclidean state space; then, a generalized version of the algorithm (13) is given by

$$\begin{cases} \dot{x} \in -\partial f(x) - \lambda - \rho(x - d - (L \otimes I_m)z) - \aleph_\mu \\ \dot{\mu} = -\mu + (\mu + g(x))^+ \\ \dot{\lambda} = -(L \otimes I_m)\lambda + (x - d - (L \otimes I_m)z) \\ \dot{z} = x - d - (L \otimes I_m)z. \end{cases}$$

Theorem 3.2: Under Assumptions 2.2–2.4, x^* is a solution of nonlinear programming problem (8) if and only if there exist μ^* , λ^* , and z^* such that $(x^*, \mu^*, \lambda^*, z^*)$ is an equilibrium point of distributed algorithm (13).

Proof: Assume that $(x^*, \mu^*, \lambda^*, z^*)$ is an equilibrium point of algorithm (13), and then, there exist $\gamma^* \in \partial f(x^*)$ and $\xi_\mu^* \in \aleph_\mu^* = \text{col}(\sum_{j=1}^{J^1} \hat{\mu}_{1j}^* \partial g_1^j(x_1^*), \dots, \sum_{j=1}^{J^n} \hat{\mu}_{nj}^* \partial g_n^j(x_n^*))$ such that

$$0 = -\gamma^* - \lambda^* - \rho(x^* - d - (L \otimes I_m)z^*) - \xi_\mu^* \quad (14a)$$

$$0 = -\mu^* + (\mu^* + g(x^*))^+ \quad (14b)$$

$$0 = -(L \otimes I_m)\lambda^* + (x^* - d - (L \otimes I_m)z^*) \quad (14c)$$

$$0 = x^* - d - (L \otimes I_m)z^* \quad (14d)$$

where $\gamma^* = \text{col}(\gamma_1^*, \dots, \gamma_n^*)$, $\xi_\mu^* = \text{col}(\sum_{j=1}^{J^1} \hat{\mu}_{1j}^* \xi_{1j}^*, \dots, \sum_{j=1}^{J^n} \hat{\mu}_{nj}^* \xi_{nj}^*)$, and $\hat{\mu}_{ij}^* = (\mu_{ij}^* + g_i^j(x_i^*))^+ \forall i, j$. Based on Assumption 2.4, we have $1_n^T L = 0$. Then, together with $x^* - d - (L \otimes I_m)z^* = 0$, it is easy to obtain $(1_n^T \otimes I_m)(x^* - d - (L \otimes I_m)z^*) = (1_n^T \otimes I_m)(x^* - d) = 0_m$. Hence, $\sum_{i=1}^n x_i^* = \sum_{i=1}^n d_i$. According to Lemma 2.6 and $\mu^* = (\mu^* + g(x^*))^+$, we have

$$g_i^j(x_i^*) \leq 0, \mu_{ij}^* \geq 0, (\mu_{ij}^*)^T g_i^j(x_i^*) = 0 \quad \forall j, i.$$

Based on $-(L \otimes I_m)\lambda^* + (x^* - d - (L \otimes I_m)z^*) = 0$, we have $(L \otimes I_m)\lambda^* = 0$, which implies that $\lambda_1^* = \lambda_2^* = \dots = \lambda_n^* = v^* \in R^m$. In addition, $-\gamma^* - \lambda^* - \rho(x^* - d - (L \otimes I_m)z^*) - \xi_\mu^* = 0$ implies

$$0 \in \partial f_i(x_i^*) + \sum_{j=1}^{J^i} \mu_{ij}^* \partial g_i^j(x_i^*) + v^* \quad \forall i.$$

From Lemma 2.5, the equilibrium point $(x^*, \mu^*, \lambda^*, z^*)$ of (13) satisfies the optimal condition, and thus, x^* is a solution of the optimization problem (8).

In contrast, assume that x^* is a solution of the optimization problem (8). By Lemma 2.5, there exist v^* and μ^* such that (9a)–(9c) hold. Define $\lambda^* = 1 \otimes v^*$, and we have $(L \otimes I_m)\lambda^* = 0$. If $\bar{s}$ satisfies $(L \otimes I_m)\bar{s} = 0$ for any $\bar{s} \in \mathbb{R}^{nm}$, then $\bar{s} \in \ker(L \otimes I_m)$. Since $\sum_{i=1}^n x_i = \sum_{i=1}^n d_i$ holds, one has $\bar{s}(x^* - d) = s(\sum_{i=1}^n x_i - \sum_{i=1}^n d_i) = 0$. One has $x^* - d \in \text{range}(L \otimes I_m)$ based on an orthogonal decomposition [32]. Therefore, there exists z^* such that $x^* - d - (L \otimes I_m)z^* = 0$, which implies that (14c) and (14d) hold. Together with (9a), we can obtain that $(x^*, \mu^*, \lambda^*, z^*)$ satisfies (14a). According to Lemma 2.6 and (9b), (14b) holds. Hence, $(x^*, \mu^*, \lambda^*, z^*)$ is an equilibrium point of the algorithm (13), and the proof is complete. $\square$

From Theorem 3.2, we only need to show that the algorithm (13) can converge to an equilibrium point, which is given as follows.

Theorem 3.3: Let Assumptions 2.2–2.4 hold. The state vector $x(t)$ under the algorithm (13) converges asymptotically to a solution of the optimization problem (8) for any initial value $(x(0), \mu(0), \lambda(0), z(0))$.

Proof: For simplicity, we just consider $m = 1$. For the system (13), using the coordinate transformation

$$\bar{x} = x - x^*, \bar{\lambda} = \lambda - \lambda^*, \bar{z} = z - z^*$$

we have

$$\begin{cases} \dot{\bar{x}} \in -h - \bar{\lambda} - \rho(\bar{x} - L\bar{z}) - (\aleph_\mu - \aleph_\mu^*) \\ \dot{\mu} = -\mu + \hat{\mu} \\ \dot{\bar{\lambda}} = -L\bar{\lambda} + (\bar{x} - L\bar{z}) \\ \dot{\bar{z}} = \bar{x} - L\bar{z} \end{cases} \quad (15)$$

where $h = \partial f(x) - \partial f(x^*)$. Define an orthogonal transformation as follows:

$$\begin{aligned} \tilde{x} &= \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{bmatrix} = [r \ R]^T \bar{x}, \tilde{\lambda} = \begin{bmatrix} \tilde{\lambda}_1 \\ \tilde{\lambda}_2 \end{bmatrix} = [r \ R]^T \bar{\lambda}, \\ \tilde{z} &= \begin{bmatrix} \tilde{z}_1 \\ \tilde{z}_2 \end{bmatrix} = [r \ R]^T \bar{z} \end{aligned}$$

where $\tilde{x}_1, \tilde{\lambda}_1, \tilde{z}_1 \in \mathbb{R}$, $\tilde{x}_2, \tilde{\lambda}_2, \tilde{z}_2 \in \mathbb{R}^{n-1}$. Then we divide the system (15) into two subsystems as follows:

$$\begin{cases} \dot{\tilde{x}}_1 \in -r^T h - \tilde{\lambda}_1 - \rho\tilde{x}_1 - r^T(\aleph_\mu - \aleph_\mu^*) \\ \dot{\mu}_1 = -\mu_1 + \hat{\mu}_1 \\ \dot{\tilde{\lambda}}_1 = \tilde{x}_1 \\ \dot{\tilde{z}}_1 = \tilde{x}_1 \end{cases} \quad (16)$$

$$\begin{cases} \dot{\tilde{x}}_2 \in -R^T h - \tilde{\lambda}_2 - \rho(\tilde{x}_2 - R^T L R \tilde{z}_2) \\ \quad - R^T(\aleph_\mu - \aleph_\mu^*) \\ \dot{\mu}_2 = -\mu_2 + \hat{\mu}_2 \\ \dot{\tilde{\lambda}}_2 = -R^T L R \tilde{\lambda}_2 + (\tilde{x}_2 - R^T L R \tilde{z}_2) \\ \dot{\tilde{z}}_2 = \tilde{x}_2 - R^T L R \tilde{z}_2. \end{cases} \quad (17)$$

Let $\eta(t) = \text{col}(\tilde{x}(t), \mu(t), \tilde{\lambda}(t), \tilde{z}(t))$. The energy function is defined as follows:

$$\begin{aligned} V(\eta) &= \frac{1}{2} \|\tilde{x}_1\|_2^2 + \frac{1}{2} \|\mu - \mu^*\|_2^2 + \frac{1}{2} \|\tilde{\lambda}_1\|_2^2 + \frac{1}{2} \|\tilde{x}_2\|_2^2 \\ &\quad + \frac{1}{2} \|\tilde{\lambda}_2\|_2^2 + \frac{\rho}{2} \tilde{z}_2^T R^T L R \tilde{z}_2 + \frac{1}{2} \|\tilde{\lambda}_2 - \tilde{z}_2\|_2^2. \end{aligned} \quad (17)$$

It is noted that the matrix $R^T L R$ is positive definite under Assumption 2.4. The dynamics (16) is denoted as Ψ , and then, the set-valued Lie derivation of V along the system (16) is considered. For any $\zeta \in \mathcal{L}_\Psi V(\eta)$, there exist $\gamma \in \partial f(x)$ and

$\xi_\mu \in \mathbf{R}_\mu$ such that

$$\begin{aligned}
 \zeta = & -\tilde{x}_1^T (r^T(\gamma - \gamma^*) + \tilde{\lambda}_1 + \rho\tilde{x}_1 + r^T(\xi_\mu - \xi_\mu^*)) \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} (\mu_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}) + \tilde{\lambda}_1 \tilde{x}_1 \\
 & - \tilde{x}_2^T (R^T(\gamma - \gamma^*) + \tilde{\lambda}_2 + \rho(\tilde{x}_2 - R^T L R \tilde{z}_2) \\
 & + R^T(\xi_\mu - \xi_\mu^*)) + \rho \tilde{z}_2^T R^T L R (\tilde{x}_2 - R^T L R \tilde{z}_2) \\
 & - \tilde{\lambda}_2^T (R^T L R \tilde{\lambda}_2 - (\tilde{x}_2 - R^T L R \tilde{z}_2)) \\
 & + (\tilde{\lambda}_2 - \tilde{z}_2)^T (-R^T L R \tilde{\lambda}_2) \\
 = & -\tilde{x}_1^T r^T(\gamma - \gamma^*) - \tilde{x}_2^T R^T(\gamma - \gamma^*) - \rho \tilde{x}_1^2 \\
 & - \rho \|\tilde{x}_2 - R^T L R \tilde{z}_2\|_2^2 - 2\tilde{\lambda}_2^T R^T L R \tilde{\lambda}_2 \\
 & - (\mathbf{x} - \mathbf{x}^*)^T (\xi_\mu - \xi_\mu^*) \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} (\mu_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}) \\
 = & -\tilde{x}_1^T r^T(\gamma - \gamma^*) - \tilde{x}_2^T R^T(\gamma - \gamma^*) - \rho \tilde{x}_1^2 \\
 & - \rho \|\tilde{x}_2 - R^T L R \tilde{z}_2\|_2^2 - 2\tilde{\lambda}_2^T R^T L R \tilde{\lambda}_2 \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} (x_i - x_i^*)(\hat{\mu}_{ij} \xi_{ij} - \mu_{ij}^* \xi_{ij}^*) \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} (\mu_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}).
 \end{aligned}$$

Now, we consider the latter two items in (18). First, the property of projection operator [33] implies that $(\mu_{ij} + g_i^j(x_i) - \hat{\mu}_{ij})(\mu_{ij} - \mu_{ij}^*) \geq 0$. The above inequality can be reformulated as $-(\hat{\mu}_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}) \leq g_i^j(x_i)(\hat{\mu}_{ij} - \mu_{ij}^*)$. Thus, one has

$$\begin{aligned}
 & - (x_i - x_i^*)(\hat{\mu}_{ij} \xi_{ij} - \mu_{ij}^* \xi_{ij}^*) - (\mu_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}) \\
 = & - (x_i - x_i^*)(\hat{\mu}_{ij} \xi_{ij} - \mu_{ij}^* \xi_{ij}^*) - (\hat{\mu}_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}) \\
 & - \|\mu_{ij} - \hat{\mu}_{ij}\|_2^2 \\
 \leq & - (x_i - x_i^*)(\hat{\mu}_{ij} \xi_{ij} - \mu_{ij}^* \xi_{ij}^*) + g_i^j(x_i)(\hat{\mu}_{ij} - \mu_{ij}^*) \\
 & - \|\mu_{ij} - \hat{\mu}_{ij}\|_2^2 \\
 = & \hat{\mu}_{ij} \left(g_i^j(x_i) + \xi_{ij}(x_i^* - x_i) \right) \\
 & + \hat{\mu}_{ij}^* \left(-g_i^j(x_i) + \xi_{ij}^*(x_i - x_i^*) \right) - \|\mu_{ij} - \hat{\mu}_{ij}\|_2^2 \\
 \leq & -\hat{\mu}_{ij} \left(g_i^j(x_i) - g_i^j(x_i^*) - \xi_{ij}(x_i^* - x_i) \right) + \hat{\mu}_{ij} g_i^j(x_i^*) \\
 & - \hat{\mu}_{ij}^* \left(g_i^j(x_i) - g_i^j(x_i^*) - \xi_{ij}^*(x_i - x_i^*) \right) - \|\mu_{ij} - \hat{\mu}_{ij}\|_2^2.
 \end{aligned}$$

The last inequality holds due to $\mu_{ij}^* g(x_i^*) = 0$. Since the function g_i^j is convex and $\hat{\mu}_{ij} g(x_i^*) \leq 0$, we have

$$- (x_i - x_i^*)(\hat{\mu}_{ij} \xi_{ij} - \mu_{ij}^* \xi_{ij}^*) - (\mu_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}) \leq 0.$$

Then, we have

$$\begin{aligned}
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} (x_i - x_i^*)(\hat{\mu}_{ij} \xi_{ij} - \mu_{ij}^* \xi_{ij}^*) \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} (\mu_{ij} - \mu_{ij}^*)(\mu_{ij} - \hat{\mu}_{ij}) \leq 0.
 \end{aligned}$$

On the other hand, it is noted that

$$\begin{aligned}
 & -\tilde{x}_1^T r^T(\gamma - \gamma^*) - \tilde{x}_2^T R^T(\gamma - \gamma^*) \\
 = & -\tilde{\mathbf{x}}^T [r \ R]^T (\gamma - \gamma^*) \\
 = & -(\mathbf{x} - \mathbf{x}^*)^T [r \ R] [r \ R]^T (\gamma - \gamma^*) \\
 = & -(\mathbf{x} - \mathbf{x}^*)^T (\gamma - \gamma^*) \leq 0.
 \end{aligned}$$

(18) Then, we have

$$\begin{aligned}
 \zeta \leq & -(\mathbf{x} - \mathbf{x}^*)^T (\gamma - \gamma^*) - \rho \tilde{x}_1^2 \\
 & - \rho \|\tilde{x}_2 - R^T L R \tilde{z}_2\|_2^2 - 2\tilde{\lambda}_2^T R^T L R \tilde{\lambda}_2 \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}_{ij} \left(g_i^j(x_i^*) - g_i^j(x_i) - \xi_{ij}(x_i^* - x_i) \right) \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}_{ij}^* \left(g_i^j(x_i) - g_i^j(x_i^*) - \xi_{ij}^*(x_i - x_i^*) \right) \\
 & + \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}_{ij} g_i^j(x_i^*) - \sum_{i=1}^n \sum_{j=1}^{J^i} \|\mu_{ij} - \hat{\mu}_{ij}\|_2^2 \leq 0.
 \end{aligned}$$

Due to the arbitrariness of ζ , one has $\max \mathcal{L}_\Psi V(\boldsymbol{\eta}) \leq 0$. Since $V(\boldsymbol{\eta})$ is nonincreasing, we have $V(\boldsymbol{\eta}(t)) \leq V(\boldsymbol{\eta}(0))$. Furthermore, $(\mathbf{0}, \boldsymbol{\mu}^*, \mathbf{0}, \mathbf{0})$ is fixed. As a result, the solution $(\tilde{\mathbf{x}}(t), \boldsymbol{\mu}(t), \tilde{\boldsymbol{\lambda}}(t), \tilde{\mathbf{z}}(t))$ is bounded. Based on Lemma 2.1, the solution of the algorithm (16) for any initial value converges to the largest weakly positively invariant set $M \in S$, where the set S is $\{\boldsymbol{\eta} : \mathbf{0} \in \mathcal{L}_\Psi V(\boldsymbol{\eta})\}$. Let $\boldsymbol{\eta}' \in M$; then, $\boldsymbol{\zeta}' = \mathbf{0} \in \mathcal{L}_\Psi V(\boldsymbol{\eta}')$. According to (19), there exist $\boldsymbol{\gamma}' \in \partial f(\mathbf{x}')$ and $\xi_{ij}' \in \partial g_i^j(x_i')$, such that

$$\begin{aligned}
 \mathbf{0} = \boldsymbol{\zeta}' \leq & -(\mathbf{x}' - \mathbf{x}^*)^T (\boldsymbol{\gamma}' - \boldsymbol{\gamma}^*) - \rho \|\tilde{x}_1'\|^2 \\
 & - \rho \|\tilde{x}_2' - R^T L R \tilde{z}_2'\|_2^2 - 2(\tilde{\lambda}_2')^T R^T L R \tilde{\lambda}_2' \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}_{ij}' \left(g_i^j(x_i^*) - g_i^j(x_i') - \xi_{ij}'(x_i^* - x_i') \right) \\
 & - \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}_{ij}'^* \left(g_i^j(x_i') - g_i^j(x_i^*) - \xi_{ij}'^*(x_i' - x_i^*) \right) \\
 & + \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}_{ij}' g_i^j(x_i^*) - \sum_{i=1}^n \sum_{j=1}^{J^i} \|\mu_{ij}' - \hat{\mu}_{ij}'\|_2^2 \leq 0.
 \end{aligned}$$

Thus we have the following relationships:

$$\rho \|\tilde{x}_2' - R^T L R \tilde{z}_2'\|_2^2 + 2(\tilde{\lambda}_2')^T R^T L R \tilde{\lambda}_2' = 0 \quad (20a)$$

$$\|\tilde{x}_1'\|^2 + \|\tilde{x}_2' - R^T L R \tilde{z}_2'\|_2^2 = 0 \quad (20b)$$

$$\sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}'_{ij} g_i^j(x_i^*) = 0 \quad (20c)$$

$$\sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}'_{ij} \left(g_i^j(x_i^*) - g_i^j(x'_i) - \xi'_{ij}(x_i^* - x'_i) \right) = 0 \quad (20d)$$

$$\sum_{i=1}^n \sum_{j=1}^{J^i} \|\mu'_{ij} - \hat{\mu}'_{ij}\|_2^2 = 0. \quad (20e)$$

According to (20b), we have

$$\tilde{x}' = \begin{bmatrix} 0 \\ R^T L R \tilde{z}'_2 \end{bmatrix}.$$

With the coordinate transformation, it is easy to derive that

$$x' - x^* = [r \ R] \tilde{x}' = [r \ R] \begin{bmatrix} 0 \\ R^T L R R^T \tilde{z}' \end{bmatrix} = L \tilde{z}'. \quad (21)$$

Moreover, combined with (20c) and (20d), we have

$$\begin{aligned} & \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}'_{ij} \left(g_i^j(x_i^*) - g_i^j(x'_i) - \xi'_{ij}(x_i^* - x'_i) \right) \\ &= - \sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}'_{ij} \left(g_i^j(x'_i) + \xi'_{ij}(x_i^* - x'_i) \right) = 0. \end{aligned} \quad (22)$$

Since $\mu' = (\mu' + g(x'))^+$ means $\hat{\mu}'_{ij} g_i^j(x'_i) = 0$, we can conclude from (22) that

$$\sum_{i=1}^n \sum_{j=1}^{J^i} \hat{\mu}'_{ij} \xi'_{ij}(x_i^* - x'_i) = 0. \quad (23)$$

Substituting (20a), (20e), (21), and (23) into (18) leads to

$$\begin{aligned} & (x' - x^*)^T (\gamma' - \gamma^*) \\ &+ \rho \|\tilde{x}'_2 - R^T L R \tilde{z}'_2\|_2^2 + 2(\tilde{\lambda}'_2)^T R^T L R \tilde{\lambda}'_2 \\ &+ \sum_{i=1}^n \sum_{j=1}^{J^i} (x'_i - x_i^*)^T (\hat{\mu}'_{ij} \xi'_{ij} - \mu_{ij}^* \xi_{ij}^*) \\ &+ \sum_{i=1}^n \sum_{j=1}^{J^i} (\mu'_{ij} - \mu_{ij}^*) (\mu'_{ij} - \hat{\mu}'_{ij}) \\ &= (x' - x^*)^T (\gamma' - \gamma^*) \\ &+ \sum_{i=1}^n \sum_{j=1}^{J^i} (x'_i - x_i^*) (\hat{\mu}'_{ij} \xi'_{ij} - \mu_{ij}^* \xi_{ij}^*) \\ &= (x' - x^*)^T \gamma' + (L \tilde{z}')^T \lambda^* + \sum_{i=1}^n \sum_{j=1}^{J^i} (x'_i - x_i^*) (\hat{\mu}'_{ij} \xi'_{ij}) \\ &= (x' - x^*)^T \gamma' = 0. \end{aligned}$$

Due to the convexity of $f(x)$, one has

$$f(x^*) - f(x') \geq (\gamma')^T (x^* - x) = 0.$$

Thus, we have $f(x^*) \geq f(x')$. Besides, x^* is a solution to the optimization problem (8), and thus, one has $f(x') \geq f(x^*)$. Then, we have $f(x') = f(x^*)$. Due to the arbitrariness of η' in the set M , the state $x(t)$ starting from any state in M is an optimal solution to the problem (8).

Next, we prove that the state $x(t)$ can globally converge to a solution of the optimization problem (8). Because $\eta(\cdot)$ is bounded, there exist $\eta_0 = (\tilde{x}_0, \mu_0, \tilde{\lambda}_0, \tilde{z}_0)$ and $\{t_k, k = 1, 2, \dots\}$ such that $\eta(t_k) = (\tilde{x}(t_k), \mu(t_k), \tilde{\lambda}(t_k), \tilde{z}(t_k)) \rightarrow (\tilde{x}_0, \mu_0, \tilde{\lambda}_0, \tilde{z}_0)$ as $k \rightarrow +\infty$. By Lemma 2.1, we have $\text{dist}(\eta(t), M) \rightarrow 0$. Thus, we conclude that $\eta_0 \in M$. Based on the aforementioned discussion, x_0 is a solution of the optimization problem (8). A new Lyapunov function $V_2(\eta)$ is defined as $V(\eta)$ by replacing the point $(0, \mu^*, 0, 0)$ in (17) with $(\tilde{x}_0, \mu_0, \tilde{\lambda}_0, \tilde{z}_0)$. By using a similar discussion procedure for $V(\eta)$ as above, we have $\max \mathcal{L}_\Psi V_2(\eta(t)) \leq 0$. Since the function V_2 is continuous, for any $\epsilon > 0$, there exists $\omega > 0$ such that $V_2(\eta(t)) \leq \epsilon$, when $\|\eta - \eta_0\| \leq \omega$. Since V_2 is monotonically nonincreasing during the time interval $[0, +\infty)$, there exists a positive integer T such that, when $t \geq t_T$, one has

$$\frac{1}{2} \|\tilde{x}(t) - \tilde{x}_0\|^2 \leq V_2(\eta(t)) \leq V_2(\eta(t_T)) \leq \epsilon.$$

Hence, $\lim_{t \rightarrow +\infty} \tilde{x}(t) = \tilde{x}_0$. The solution $\tilde{x}(t)$ of the algorithm (16) can globally converge to the set M since V_2 is radially unbounded with respect to $\tilde{x}(t)$. Then, the state vector $x(t)$ of the algorithm (13) can globally converge to an optimal solution of problem (8) based on the affine and static transformation. $\square$

IV. SIMULATION RESULTS

In this section, we provide a numerical example to demonstrate that the economic dispatch problem of energy–water nexus (1)–(4) can be solved by the proposed algorithm (13). Assume that the cost functions $C_{pi}(x_{pi})$, $C_{wj}(x_{wj})$, and $C_{ck}(x_{cpk}, x_{cwk})$ have a quadratic form with their respective production variables

$$\begin{aligned} C_{pi}(x_{pi}) &= \frac{1}{2} A_{pi,11} x_{pi}^2 + B_{pi,1} x_{pi} + K_{pi} \\ C_{wj}(x_{wj}) &= \frac{1}{2} A_{wj,11} x_{wj}^2 + B_{wj,1} x_{wj} + K_{wj} \\ C_{ck}(x_{cpk}, x_{cwk}) &= \frac{1}{2} A_{ck,11} x_{cpk}^2 + B_{ck,1} x_{cpk} \\ &\quad + \frac{1}{2} A_{ck,22} x_{cwk}^2 + B_{ck,2} x_{cwk} \\ &\quad + \frac{1}{2} A_{ck,12} x_{cpk} x_{cwk} + K_{ck}. \end{aligned}$$

Moreover, it is assumed that there are two power plants, two reverse osmosis water plants, and two coproduction desalination facilities in the network. The undirected and connected communication graph of the power–water network is illustrated in Fig. 2, and we adopt the period communication strategy to implement the continuous algorithm (13) (i.e., every agent exchanges the information with its neighbors synchronously at a time interval). The initial values are chosen as follows: $x(0) = 0$, $\mu(0) = 0$, $\lambda(0) = 0$, and $z(0) = 0$. The algorithm (13) is implemented

TABLE I
PARAMETERS OF PLANTS' CONSTRAINTS

Plants	Index	Max generating power capacity (MW)	Min generating power capacity (MW)	Max production water capacity (m^3/h)	Min production water capacity (m^3/h)	Min product ratio	Max product ratio
power	i_1	200	0	0	0	—	—
power	i_2	300	0	0	0	—	—
water	j_1	0	0	250	0	—	—
water	j_2	0	0	335	0	—	—
coproduction	k_1	800	160	200	30	4	9
coproduction	k_2	600	120	150	23	3	6

TABLE II
COST PARAMETERS AND RESOURCE DEMAND AT $t = 0\text{--}200\text{~s}$

The parameters of power plants					The parameters of water plants				
Index	$A_{pi,11}$	$B_{pi,1}$	K_{pi}	d_{pi}	Index	$A_{wj,11}$	$B_{wj,1}$	K_{wj}	d_{wj}
i_1	0.0575	1.2	5	196	j_1	0.0225	1	7	260
i_2	0.0475	1.75	10	218	j_2	0.03834	3.25	5.8	40
The parameters of coproduction plants									
Index	$A_{ck,11}$	$A_{ck,12}$	$A_{ck,22}$	$B_{ck,1}$	$B_{ck,2}$	K_{ck}	d_{cpk}	d_{cwk}	
k_1	0.04433	0.03546	0.027093	1.106	0.426	57	30	114	
k_2	0.037881	0.026305	0.03261	1.475	5.901	57	220	225	

TABLE III
COST PARAMETERS AND RESOURCE DEMAND AT $t = 200\text{--}400\text{~s}$

The parameters of power plants					The parameters of water plants				
Index	$A_{pi,11}$	$B_{pi,1}$	K_{pi}	d_{pi}	Index	$A_{wj,11}$	$B_{wj,1}$	K_{wj}	d_{wj}
i_1	0.0775	3.2	5	105	j_1	0.0425	3.0	7	143
i_2	0.0375	2.75	10	226	j_2	0.05834	2.25	5.8	8
The parameters of coproduction plants									
Index	$A_{ck,11}$	$A_{ck,12}$	$A_{ck,22}$	$B_{ck,1}$	$B_{ck,2}$	K_{ck}	d_{cpk}	d_{cwk}	
k_1	0.06433	0.04546	0.057093	3.106	0.726	57	197	185	
k_2	0.047881	0.036305	0.02261	3.475	8.901	57	237	204	

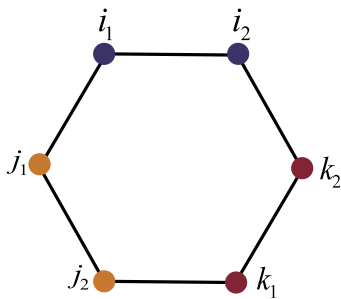


Fig. 2. Topology structure of communication network.

to solve the dispatch problem with the data given in Tables I–IV. At $t = 200\text{~s}$ and $t = 400\text{~s}$, the cost and resource data are changed. The results obtained from the numerical simulation are given in Figs. 3–6.

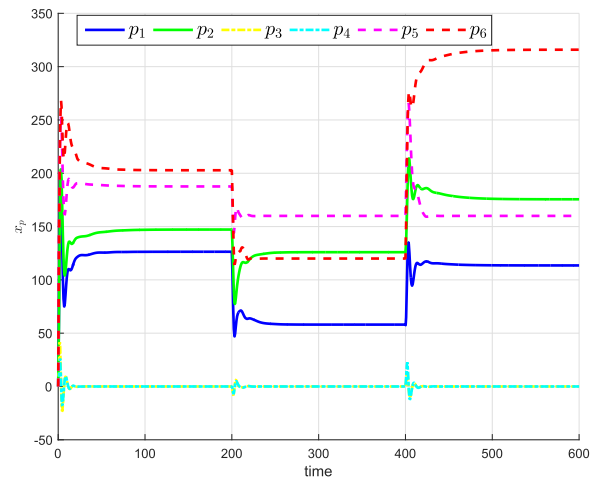


Fig. 3. Decision variables of power plants.

TABLE IV
COST PARAMETERS AND RESOURCE DEMAND AT $t = 400 \text{ s} - +\infty$

The parameters of power plants					The parameters of water plants				
Index	$A_{pi,11}$	$B_{pi,1}$	K_{pi}	d_{pi}	Index	$A_{wj,11}$	$B_{wj,1}$	K_{wj}	d_{wj}
i_1	0.0975	0.2	5	146	j_1	0.0615	1.0	7	176
i_2	0.0575	2.15	10	147	j_2	0.07234	4.25	5.8	90
The parameters of coproduction plants									
Index	$A_{ck,11}$	$A_{ck,12}$	$A_{ck,22}$	$B_{ck,1}$	$B_{ck,2}$	K_{ck}	d_{cpk}	d_{cwk}	
k_1	0.08133	0.06546	0.037093	1.106	2.726	57	24	148	
k_2	0.027881	0.056305	0.07261	1.475	5.901	57	147	26	

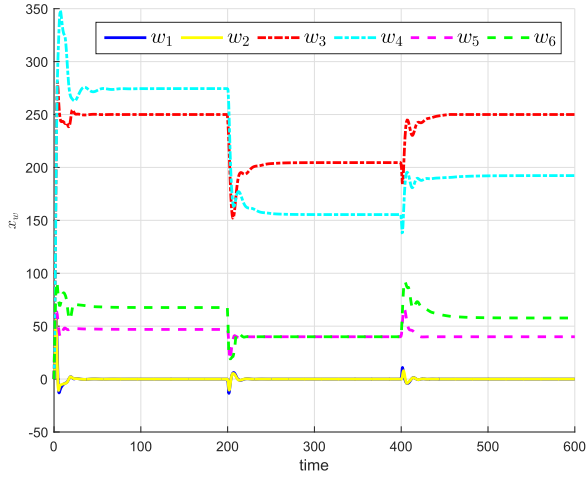


Fig. 4. Decision variables of water plants.

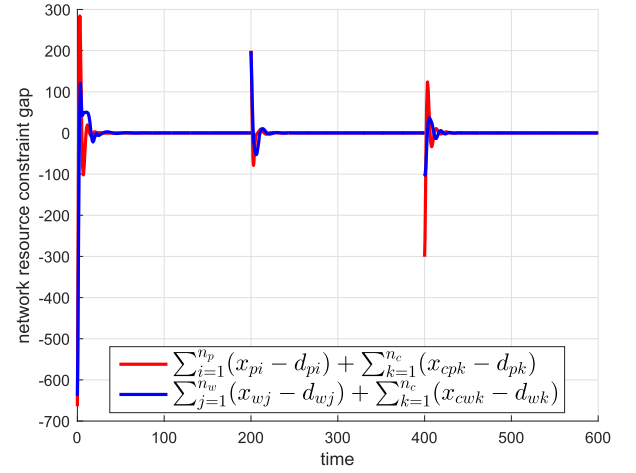


Fig. 6. Evolution of the supply-demand constraint.

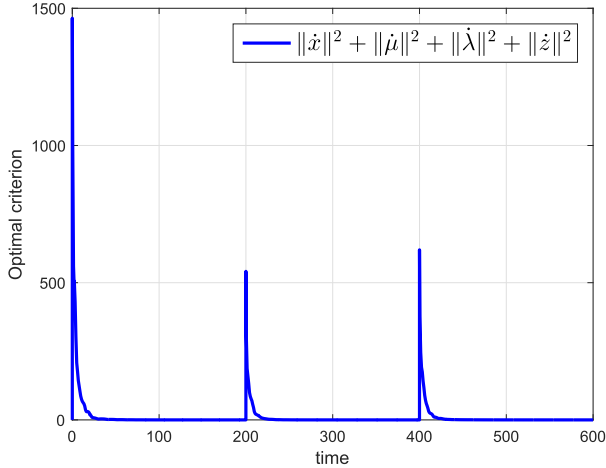


Fig. 5. Evolution of an optimization criterion.

Figs. 3 and 4 show the dynamic evolution of the decision variables about power plants and water plants, respectively. In Fig. 5, the variable $\|\dot{x}\|^2 + \|\dot{\mu}\|^2 + \|\dot{\lambda}\|^2 + \|\dot{z}\|^2$ gradually converges to zero, implying that the decision variables can gradually reach the equilibrium point. Based on Theorem 3.2, the decision variables can converge to the optimal point. Therefore, the designed algorithm (13) is still effective even when the data/configuration

changes. At the last stage, the decision variables about the power and water converge to the optimal output (113.54, 175.57, 0, 0, 160, 315.89) and (0, 0, 250, 192, 28, 40, 57.72), respectively. Fig. 6 shows that the supply-demand balance can be kept after some time.

V. CONCLUSION

In this study, a nonlinear programming method has been adopted to formulate an economic dispatch problem for a hybrid energy system with power, water, and coproduction plants. Subsequently, a continuous time distributed algorithm has been proposed for an economic dispatch problem of the hybrid system starting from any initial allocation. The proposed algorithm can effectively solve the economic dispatch problem with local nonlinear inequality constraints. To accomplish this, a complementary equivalent condition has been adopted to design the distributed algorithm. The proposed algorithm has been tested on an economic dispatch problem of a power-water network, and the simulation results indicated that the designed algorithm converges to an optimal solution. In our future research, an economic dispatch problem of the power-water network with electricity storage devices and water storage facilities would be considered.

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Peng Li received the M.S. and Ph.D. degrees in control science and engineering from the University of Electronic Science and Technology of China, Chengdu, China, in 2016 and 2021, respectively.

From 2019 to 2020, he was a Visiting Student with the Dipartimento di Elettronica, Informazione e Bioingegneria, Politecnico di Milano, Milano, Italy, under the grant of the China Scholarship Council. His research interests include distributed optimization, multiagent systems, and smart grids.



Jiangping Hu (Senior Member, IEEE) received the B.S. degree in applied mathematics and the M.S. degree in computational mathematics from Lanzhou University, Lanzhou, China, in 2000 and 2004, respectively, and the Ph.D. degree in modeling and control of complex systems from the Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China, in 2007.

He has held various positions with the KTH Royal Institute of Technology, Stockholm, Sweden; City University of Hong Kong, Hong Kong; Sophia University, Tokyo, Japan; University of Western Sydney, Penrith, NSW, Australia. He is currently a Professor with the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China. His current research interests include multiagent systems, social dynamics, and sensor networks.

Dr. Hu was an Associate Editor for *KYBERNETIKA* and an Associate Editor for *Journal of Systems Science and Complexity*.



Li Qiu (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Toronto, Toronto, ON, Canada, in 1990.

After briefly working in the Canadian Space Agency, the Fields Institute for Research in Mathematical Sciences (Waterloo, ON, Canada), and the Institute of Mathematics and its Applications (Minneapolis, MN, USA), he joined the Hong Kong University of Science and Technology, Kowloon, Hong Kong, in 1993, where he is currently a Professor of Electronic and Computer Engineering. His research interests include control theory for systems and networks.

Dr. Qiu was an Associate Editor for IEEE TRANSACTIONS ON AUTOMATIC CONTROL and an Associate Editor for *Automatica*. He was the General Chair of the 7th Asian Control Conference in 2009. He was a Distinguished Lecturer from 2007 to 2010 and a member of the Board of Governors in 2012 and 2017 of the IEEE Control Systems Society. He is a member of the steering committees of the Asian Control Association and the International Symposium of Mathematical Theory of Networks and Systems. He is the founding chairperson of the Hong Kong Automatic Control Association, having served the term 2014–2020. He is a Fellow of the International Federation for Automatic Control.



Yiyi Zhao received the master's degree in region economics from Yunnan Minzu University, Kunming, China, in 2007, and the Ph.D. degree in management science and engineering from the School of Economics and Management, University of Electronic Science and Technology of China, Chengdu, China, in 2014.

She is currently an Associate Professor with the School of Business Administration, Faculty of Business Administration, Southwestern University of Finance and Economics, Chengdu. Her research interests include modeling and analysis of complex systems.



Bijoy Kumar Ghosh (Fellow, IEEE) received the Ph.D. degree in engineering from Harvard University, Cambridge, MA, USA, in 1983.

From 1983 to 2007, he was with the Department of Electrical and Systems Engineering, Washington University, St. Louis, MO, USA. He is currently the Dick and Martha Brooks Regents Professor of Mathematics and Statistics with Texas Tech University, Lubbock, TX, USA, and also a Professor with the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China. His research interests include biomechanics, cyberphysical systems, and control problems in rehabilitation engineering.

Dr. Ghosh became a Fellow of the International Federation on Automatic Control in 2014.